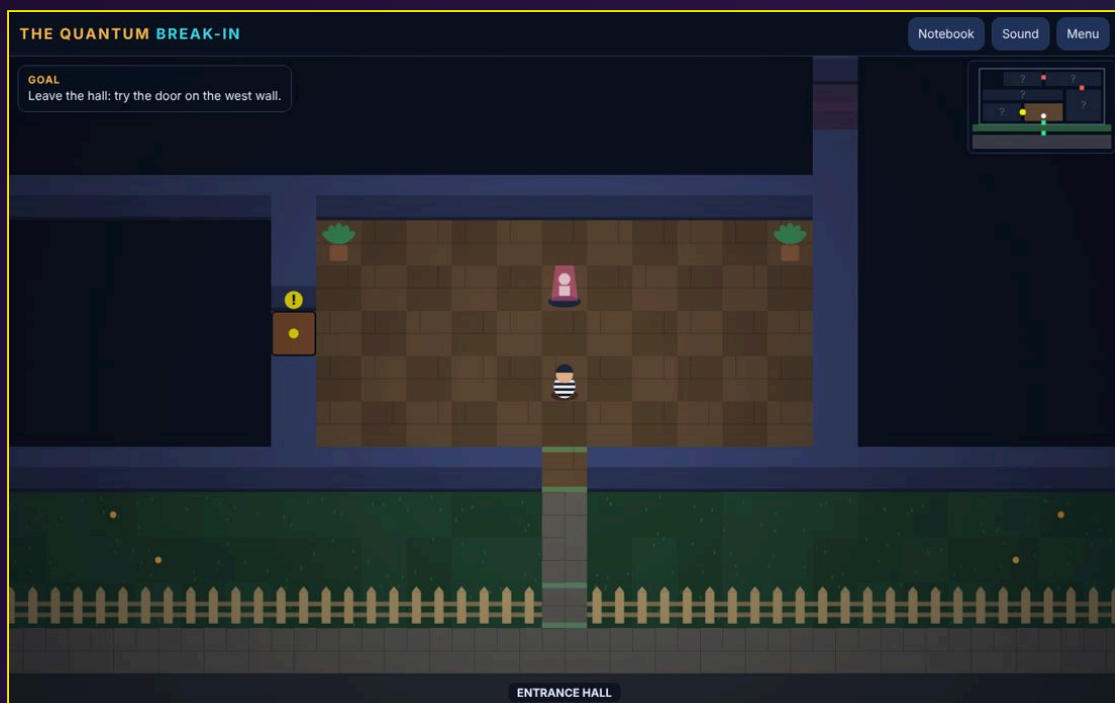


THE QUANTUM BREAK-IN

PLAYER INSTRUCTIONS

Finn is broke and about to rob the richest house on Marigold Lane. It belongs to Professor Quark, a mad scientist who protects everything with quantum locks. Finn's childhood friend Maya, a quantum physicist, is on the phone. Walk through the house, and she will teach you the physics you need to open every lock.

You do not need to know any quantum physics to play. Nothing in the game can be failed.



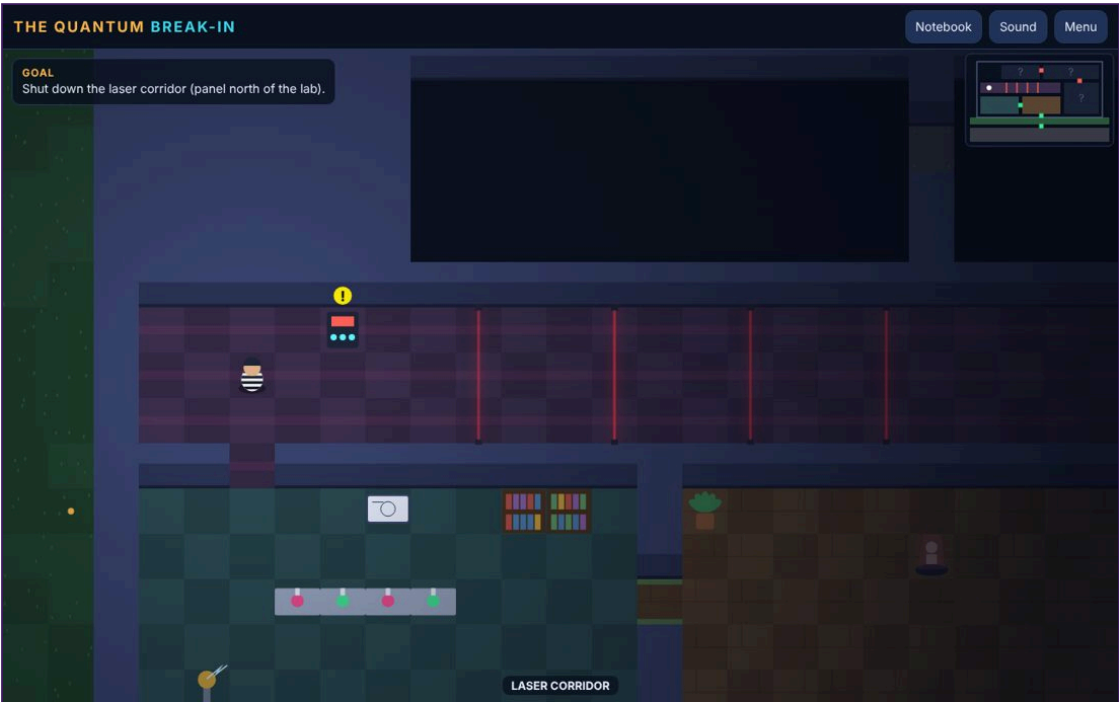
1. STARTING THE GAME

1. Open **index.html** in a web browser (double-click the file), or open the web link you were given. Chrome, Edge, Firefox and Safari all work. No installation or internet connection is needed for the file.
 2. Press **Start the break-in**. If you have played before, the button says **Continue**, and **New game** starts over.
 3. Sound starts after your first click or key press. The **Sound** button at the top mutes everything.
- Your progress saves automatically in the browser on that device. Private or incognito windows may not keep the save.

2. CONTROLS

Action	Keyboard	Mouse or touch
Move Finn	W A S D or the arrow keys	Click or tap the spot you want to walk to
Examine, read or open something	E or Space when standing next to it	Click or tap the object, or the round button at the bottom right
Continue dialogue	Space or Enter	Click or tap the dialogue box
Close a window	Esc	The Close button

Route through the house. Street → garden gate → front door → entrance hall → lab (door on the hall's west wall) → laser corridor (opening in the lab's north wall) → library (east end of the corridor) → study (door on the library's north side, needs the keycard) → vault (west end of the study, needs the crystal).



The house view: goal at top left, house plan at top right, an amber "!" over the control panel, and the laser beams blocking the corridor.

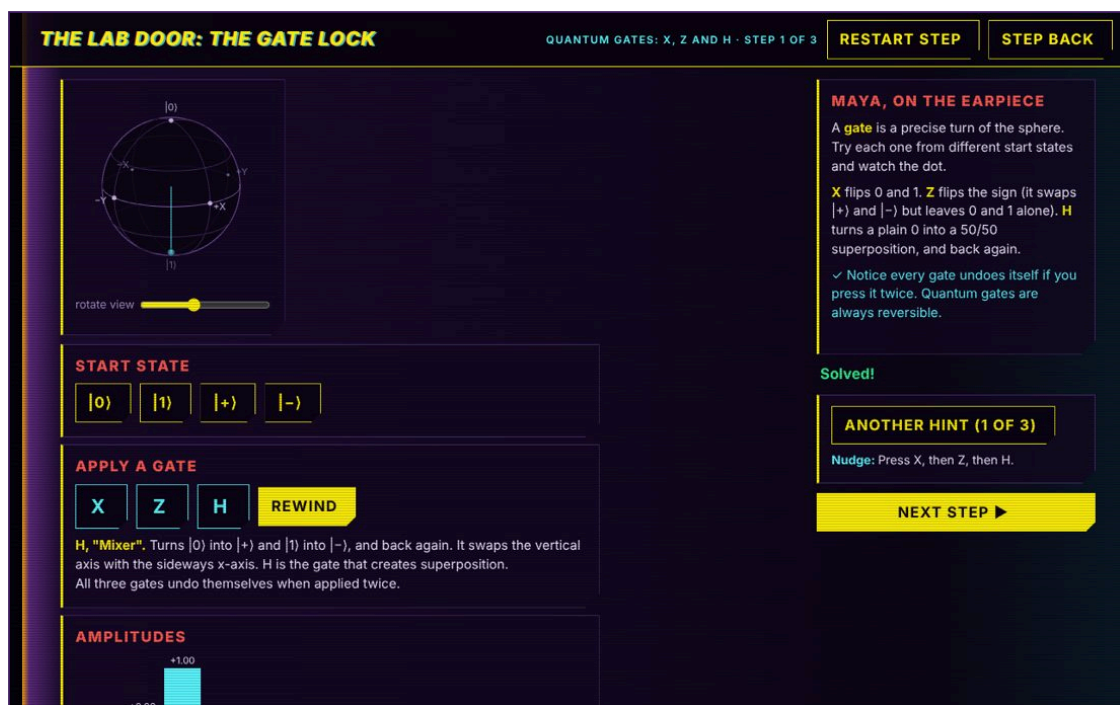
3. READING THE SCREEN

In the house

- **Goal** (top left): what to do next.
- **House plan** (top right): rooms appear once you enter them; unvisited rooms show a "?". Red squares are locked doors, green are open. The white dot is you and the blinking amber dot is your goal.
- **Amber "!"** above an object: a lock or device you have not dealt with yet.
- **Pale "?"**: one of the professor's notes. There are three; they are optional and each explains an idea.
- **White "E"**: you are close enough to interact.
- **Notebook** (top bar): a summary of every idea you have learned so far.
- **Menu** (top bar): controls reminder and Start over.

In a mini-game

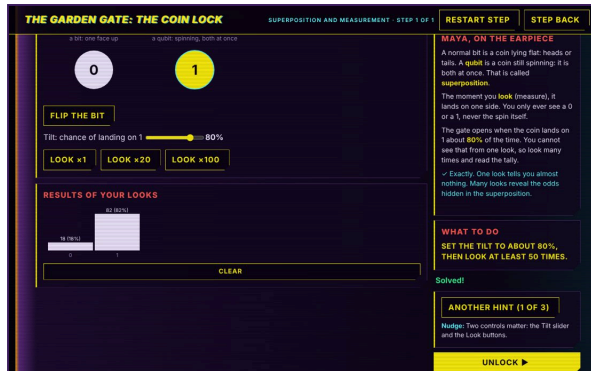
- **Left side**: the puzzle itself.
- **Maya, on the earpiece** (right): the explanation of the idea. Read this first. When you solve the step, her summary appears in cyan.
- **Ask Maya for a hint**: three levels. The first is a nudge, the second gives the method, the third is a full walkthrough. Hints cost nothing.
- **Restart step**: resets the current step with fresh values.
- **Step back**: leaves the mini-game. You keep your place and can return at any time.
- **Next step / Unlock**: appears when the step is solved.



A mini-game screen: the puzzle on the left, Maya's explanation and the hint button on the right.

4. THE SEVEN LOCKS

Each lock teaches one idea. The steps below tell you what the game wants; the third hint in the game always gives the exact moves.

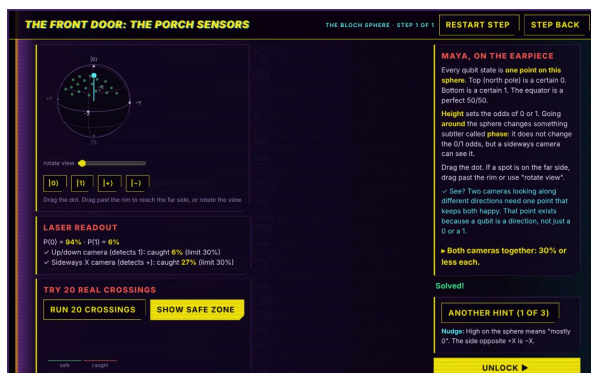


LOCK 1 · GARDEN GATE

The Coin Lock · superposition and measurement

A qubit is like a coin still spinning: both sides at once. Looking at it makes it land on one side.

- Drag the **Tilt** slider to about 80%.
- Press **Look ×100**. The gate opens when the tally shows roughly 80% ones over at least 50 looks. If the tally is unlucky, look again.

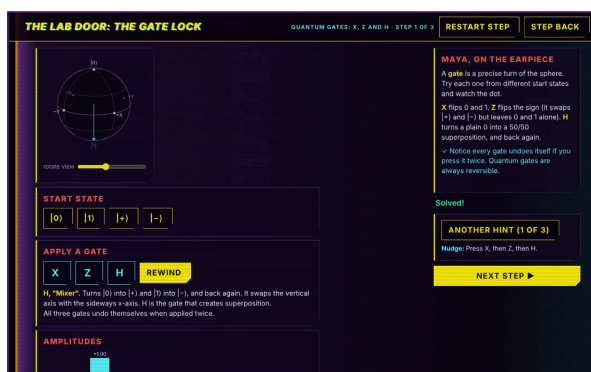


LOCK 2 · FRONT DOOR

The Porch Sensors · the Bloch sphere

Every qubit state is one point on a sphere. Height sets the odds of 0 or 1; the angle around the sphere is the phase.

- Drag the dot to complete five tasks in turn; the current task is shown at the bottom of Maya's panel.
- To reach the far side, drag past the rim or use **rotate view**. The $|0\rangle$ $|1\rangle$ $|+\rangle$ $|-\rangle$ buttons jump to four special points.
- **Show safe zone** shades the allowed area for the camera tasks.

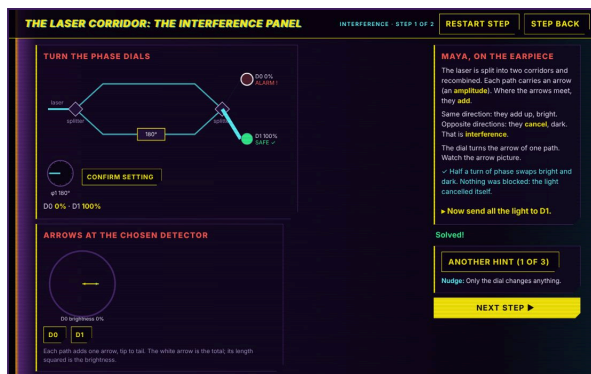


LOCK 3 · LAB DOOR

The Gate Lock · gates X, Z and H

Gates are precise turns of the sphere. X flips 0 and 1, Z flips the sign, H creates and removes superposition.

- **Step 1:** press X, Z and H once each and watch the dot.
- **Steps 2 and 3:** click gate cards to place them on the wire, then press **Run**. Click a placed card to remove it. The amber ring is the target.
- Step 3 has no X card: build a flip from H and Z.

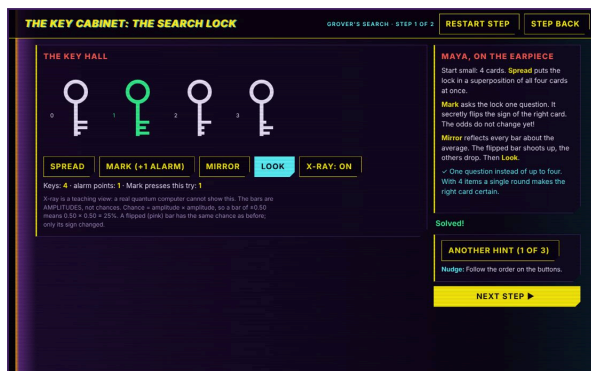


LOCK 4 · LASER CORRIDOR

The Interference Panel · interference

Light that takes several paths adds up like arrows. Aligned arrows are bright; opposite arrows cancel.

- Drag a dial to change the phase of one path. You only ever need 0° and 180°.
- Send all the light to the green SAFE detector. Click a detector name (D0 to D3) to see its arrows.
- If the light is already where it should be, press **Confirm setting**.

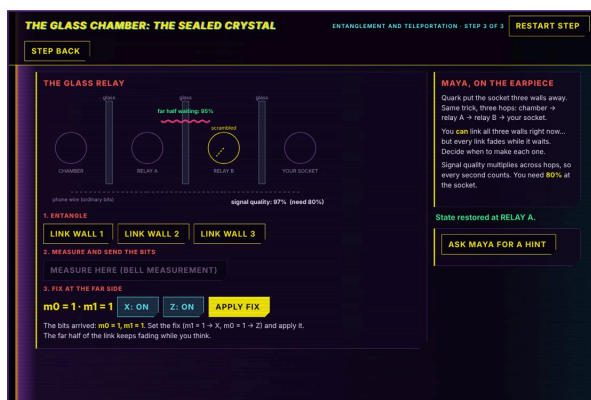


LOCK 5 · LIBRARY

The Key Cabinet · Grover's search

Find one right key among many with far fewer questions than trying them one by one.

- **Step 1 (4 keys):** press Spread, Mark, Mirror, Look, in that order.
- **Step 2 (8, then 16 keys):** press Spread, then **Grover round** the right number of times, then Look. Too few or too many rounds lowers the chance. Use 2 rounds for 8 keys and 3 for 16.
- The bars show *amplitudes*. The chance of a key is its amplitude multiplied by itself, so a bar of 0.50 means 25%.

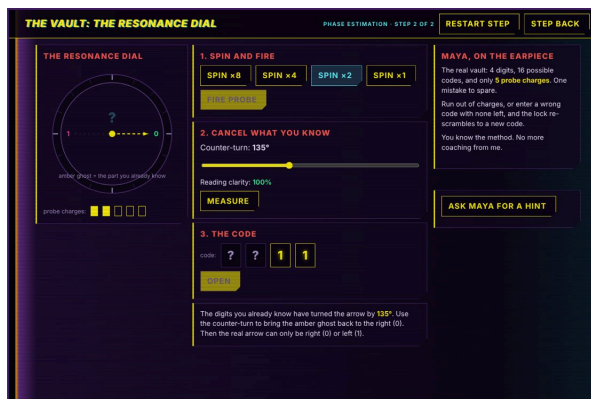


LOCK 6 · STUDY

The Glass Chamber · entanglement and teleportation

Move the crystal's state through glass without carrying it.

- **Step 1:** click the H card, then the CNOT card, then **Look ×100 at both**.
- **Steps 2 and 3 (the relay):** for each wall press **Link**, then **Measure here**, and wait for the two bits to arrive. Then set the fix: switch **X** on if $m1 = 1$, switch **Z** on if $m0 = 1$, and press **Apply fix**.
- Links fade from the moment they are made. With three walls, make each link only when the state is next to that wall. You need 90% quality for one wall and 80% for three. If it arrives too blurry, press **Run it again**.



LOCK 7 · VAULT

The Resonance Dial · phase estimation

The code is a hidden angle written in binary. Each clean probe reads one digit.

- Choose a spin, press **Fire probe**, line up the amber ghost with the green 0 using the **Counter-turn** slider until clarity reads 100%, then press **Measure**.
- Always use the **longest spin first** (×4 on the practice dial, ×8 on the real one), then work down.
- If several ghosts appear, you skipped a longer spin: that reading is a coin flip and is marked with "?".
- The real vault has 5 probe charges for 4 digits. If you run out, the lock re-scrambles and you start that step again.
- When every digit is read, press **Open**.

5. THE ENDING

Inside the vault is a sealed steel box. Walk up to it and open it to finish the story. Afterwards you can keep exploring the house, reread the notebook, or start again from the Menu.

6. TIPS AND TROUBLESHOOTING

If...	Try this
You do not know where to go	Read the Goal box and follow the blinking amber dot on the house plan.
A door says it is locked	The study needs the keycard from the library cabinet. The vault needs the crystal from the glass chamber.
You are stuck in a mini-game	Press Ask Maya for a hint up to three times. The third hint lists the exact moves.
A puzzle is in a muddle	Press Restart step .
There is no sound	Click once anywhere in the game, check the Sound button does not say "Muted", and check your device volume.
Your progress is gone	Saves are kept by the browser on one device. Clearing site data, using a private window, or switching browser or device starts a new game.
Finn will not walk to where you tapped	The spot is behind a locked door or the laser beams. Open that lock first.

The puzzles use a real quantum simulator, so results that depend on chance (looks, measurements) can differ each time, exactly as they would on real hardware.